

Introduction to microbiome data for colon model experiments

Bioinformatics 101 for colon model experiments



Petra Polakovicova
September 2026

What we'll cover

1. What data does a bioinformatician work with
2. Methodological comparison of 16S and shotgun whole-metagenome sequencing
3. Basics of the bioinformatics pipeline
4. Quick demonstration (Galaxy)

What we'll cover

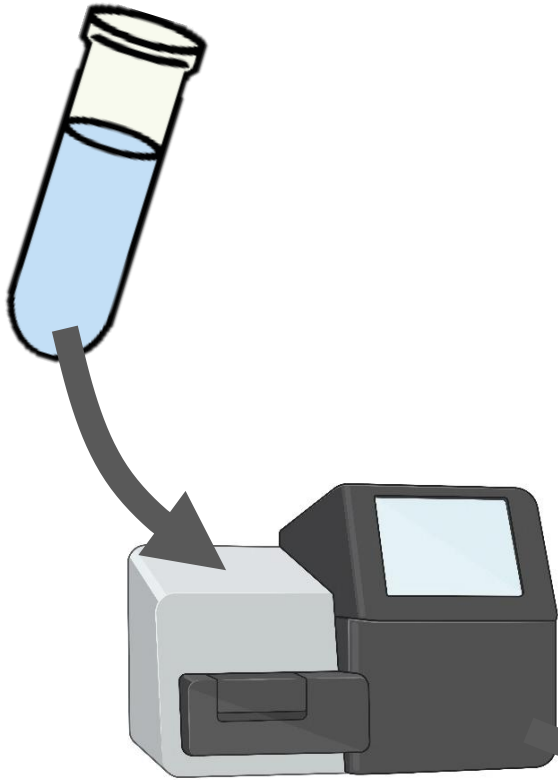
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Lecture goal: to learn the basic bioinformatics steps and understand what "data processing & analysis" really means in microbiome research

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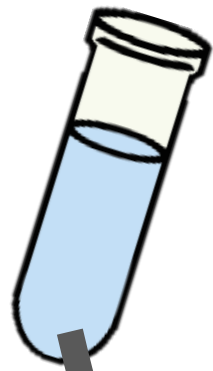
What data does a bioinformatician work with



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EGGGGGGGGGGGFFFFFFDFFFFFFFFADFFFFFFFFD-
;/9FFFFFFFAFFBCFFAFAEFFFFFFFBBFBFB/FFFFFFD
.....
.....

	S1	S2	S3	S6	S7	...	S120
Taxon 1	0	7	348	0	0	0	364
Taxon 2	10	5	46	6	47	0	62
Taxon 3	8	10	0	42	0	60	0
Taxon 4	6	0	0	95	96	668	0
Taxon 5	0	8	79	0	0	35	34
...	2	0	3	0	12	0	0
Taxon 978	450	0	55	5	32	0	12

What data does a bioinformatician work with



Sequencing output:
FASTQ file → millions of
short text strings (reads)
made up of A/C/G/T +
read quality scores

[illegible]

Abundance table

	S1	S2	S3	S6	S7	...	S120
Taxon 1	0	7	348	0	0	0	364
Taxon 2	10	5	46	6	47	0	62
Taxon 3	8	10	0	42	0	60	0
Taxon 4	6	0	0	95	96	668	0
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...	2	0	3	0	12	0	0
Taxon 978	450	0	55	5	32	0	12

What data does a bioinformatician work with

FASTQ FILE

Header

Sequence

Quality

```
@HWI-ST227:389:C4WA2ACXX:7:1204:2272:59979  
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+  
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+  
12BAFB?A:3<AE1@<FF;1*@EG*)?0?DBD>9BF9B*?#####  
@HWI-ST227:389:C4WA2ACXX:8:2208:2467:44624  
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+  
BDDA:DB?2AA@FC>F?EEGC<FED>GFD;?GBB?<?F99*/9?9?
```

4 rows per sequencing read:

- **header** starting with **@**, it can include information from the sequencing run
- **sequence** content
- spacer **(+)**
- **quality** encoded using ASCII characters (PHRED SCORE = Q)

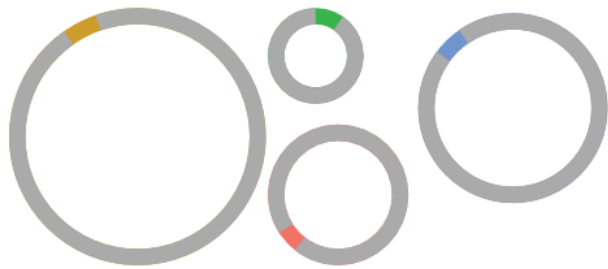
Q = 10 ('+') -> probability of incorrect base call (0.1) -> base call accuracy 90%

Q = 20 ('5') -> probability of incorrect base call (0.01) -> base call accuracy 99%

Q = 30 ('?') -> probability of incorrect base call (0.001) -> base call accuracy 99.9%

What data does a bioinformatician work with

16S rRNA amplicon sequencing

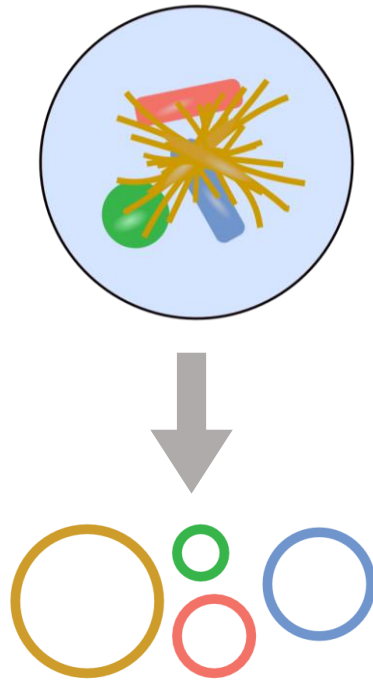


Multiple copies of fragments
from 1 target gene

Shotgun metagenomics sequencing



Short sequence
fragments from "all" DNA



What data does a bioinformatician work with

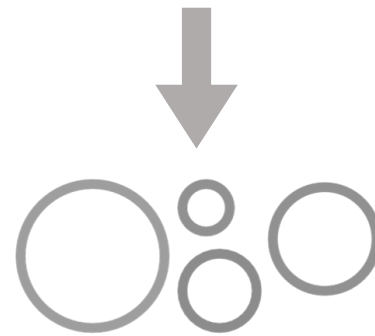
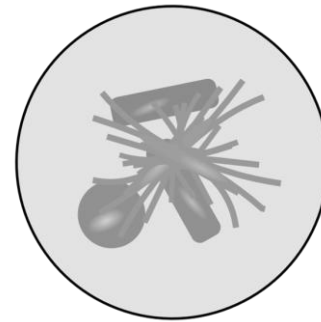
16S rRNA amplicon sequencing

1 sample ~ 20–50MB
(10 average novels'
worth of text)



Shotgun metagenomics sequencing

1 sample ~ 3–10GB
(13,000 average novels'
worth of text)



What we'll cover

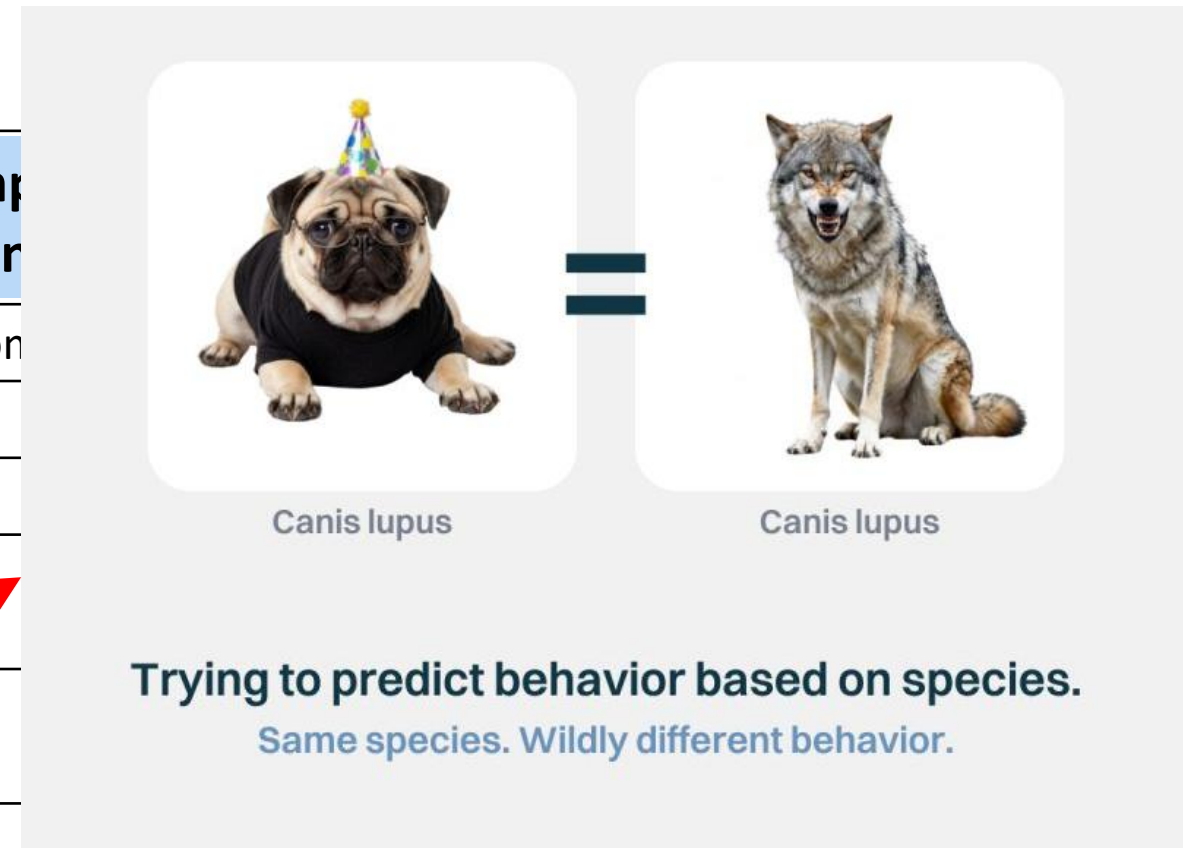
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16S vs. WMGS data

What can it tell you?	16S rRNA amplicon sequencing	Shotgun metagenomics sequencing
Best naming precision	Genus (species sometimes)	Species (strain sometimes)
Bacteria & archaea	✓	✓
Viruses & fungi	✗	✓
Who's there & how much (relative abundance)	✓	✓
Rebuild whole genomes from the mix (MAGs)	✗	✓
What the microbes can do (genes, pathways)	✗ (prediction only)	✓
Data per sample	~20–50 MB (cheap)	~3–10 GB (expensive)

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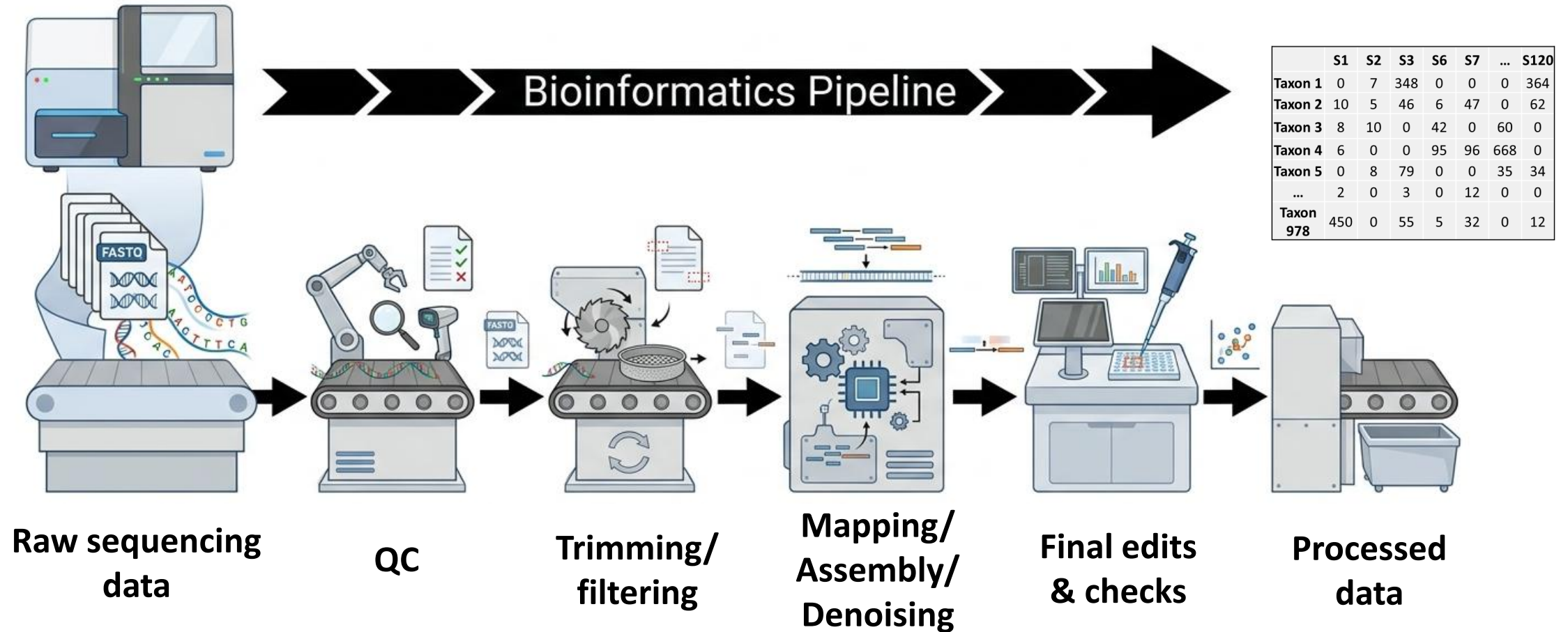


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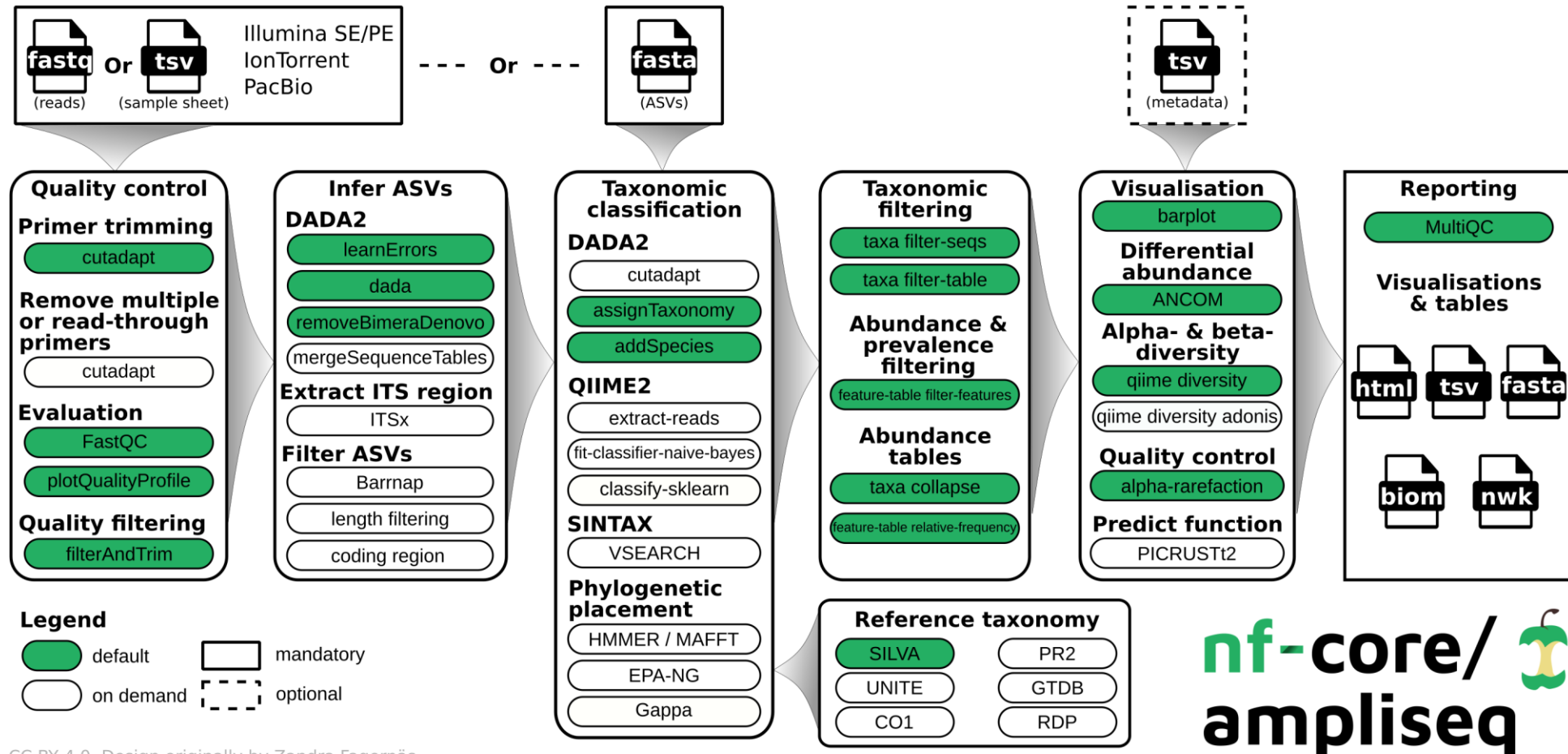
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Basics of the bioinformatics pipeline

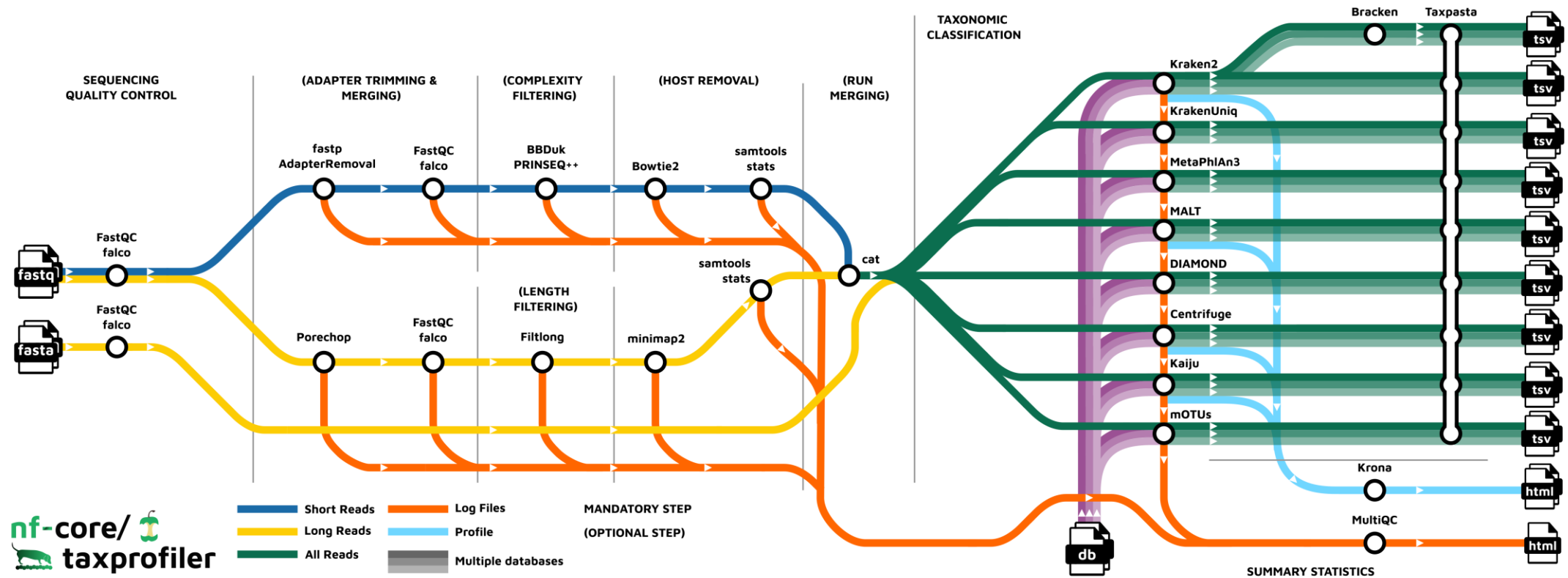
Production line in a factory



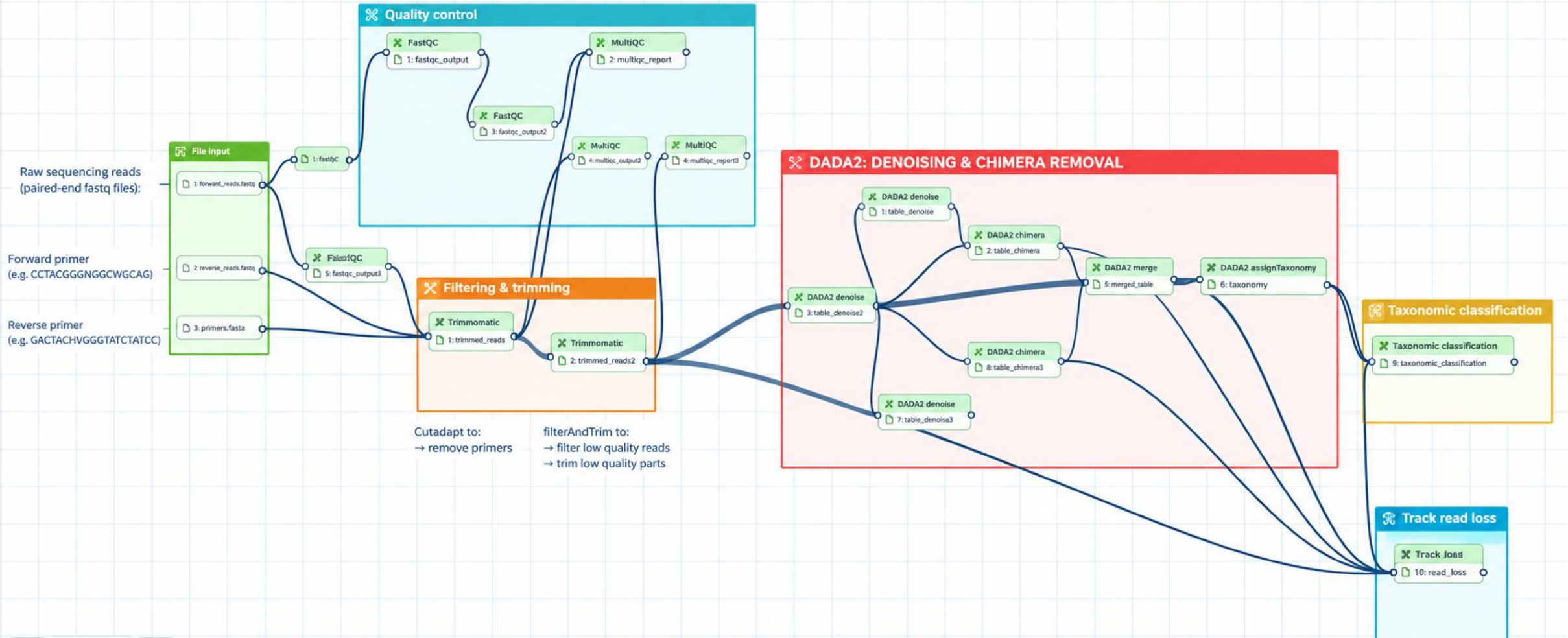
Bioinformatics pipeline - examples



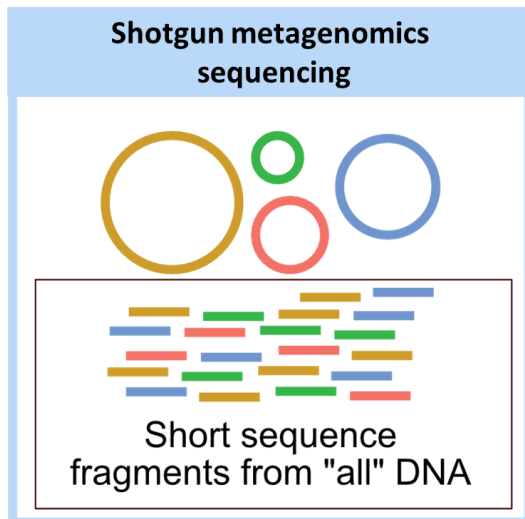
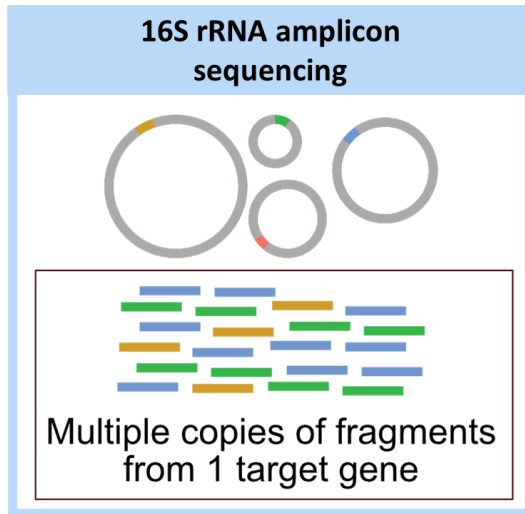
Bioinformatics pipeline - examples



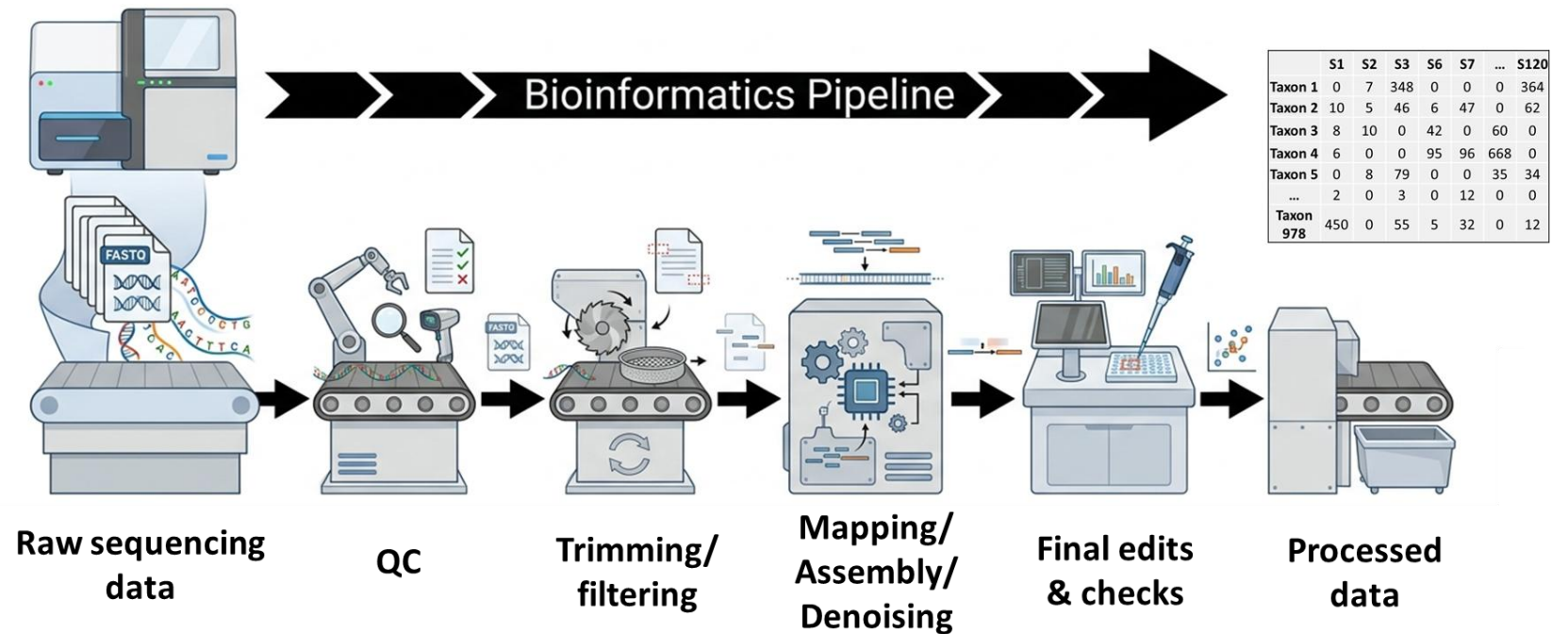
Bioinformatics pipeline - examples



Basics of the bioinformatics pipeline



?

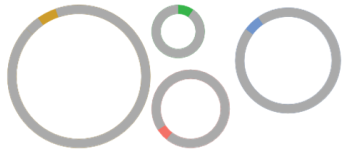


Quality control

We have reads, but are they trustworthy?

Basically the same across any type of sequencing data.

16S rRNA amplicon sequencing

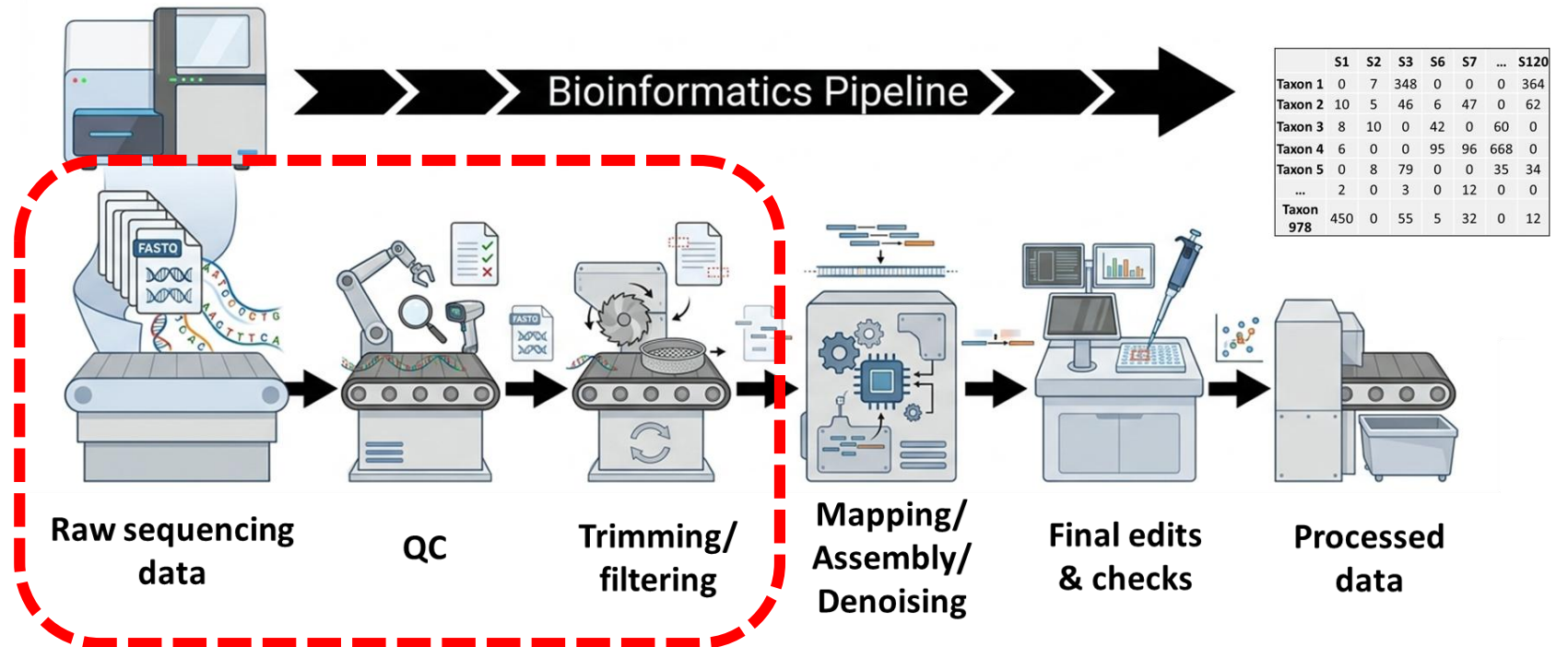


Multiple copies of fragments from 1 target gene

Shotgun metagenomics sequencing



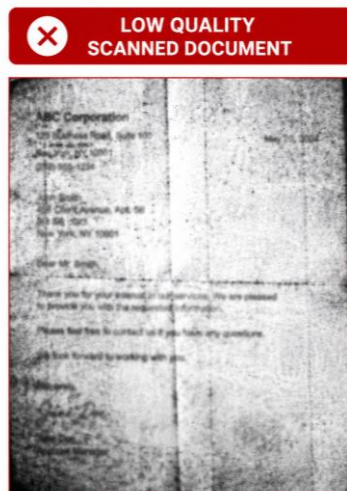
Short sequence fragments from "all" DNA



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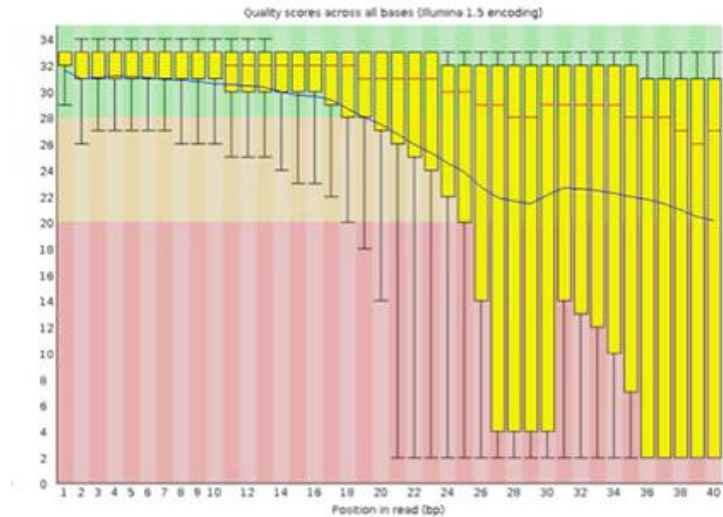
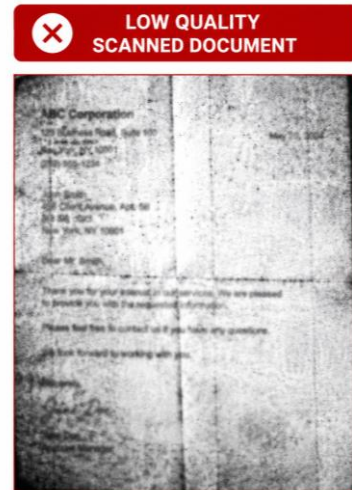
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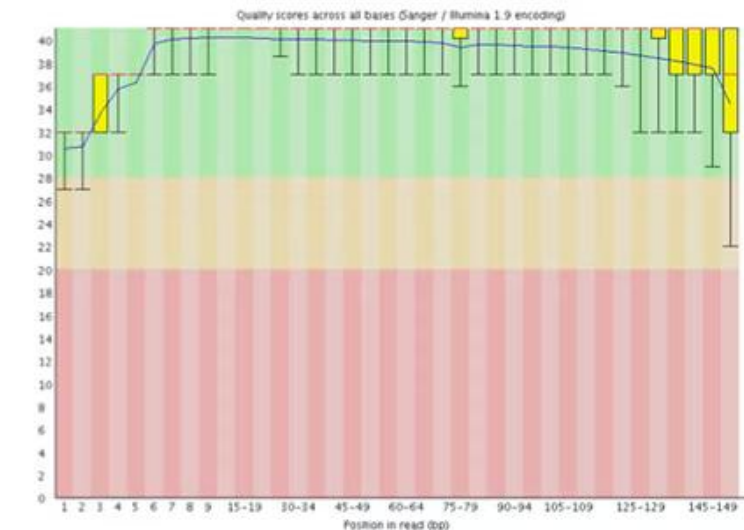
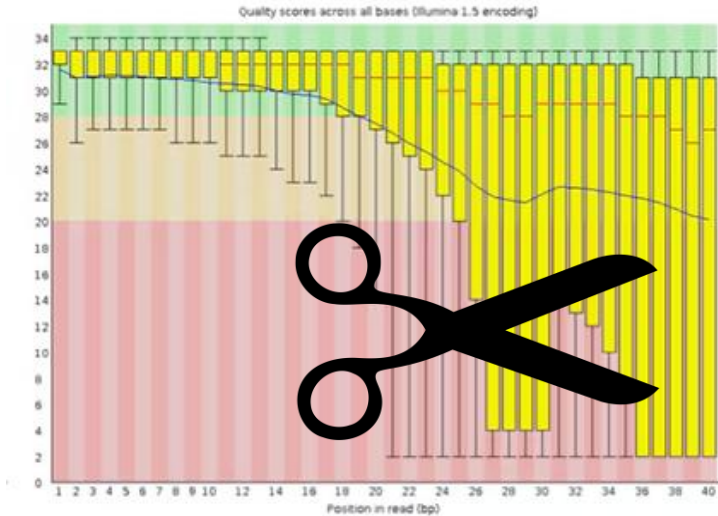
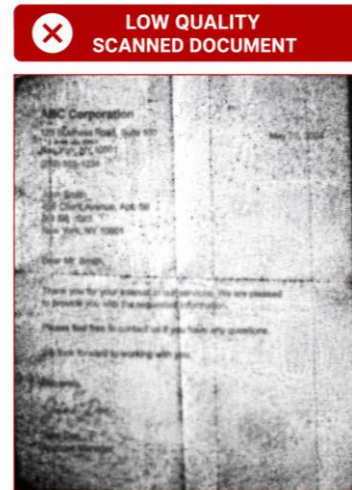
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CCAGCAGCCGCGGTAATACGGAGGATTCAAGCGTTATCCGGATT
TATTGGGTTTAAAGGGTGCGTAGGCGGTCGGG
.....
```



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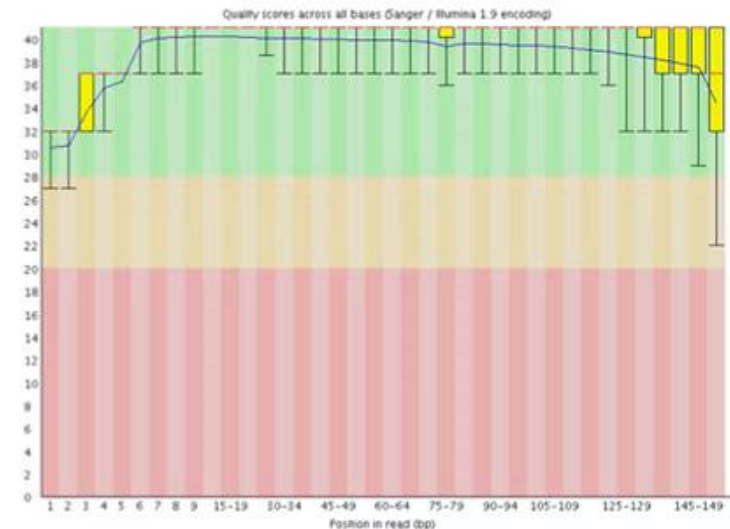
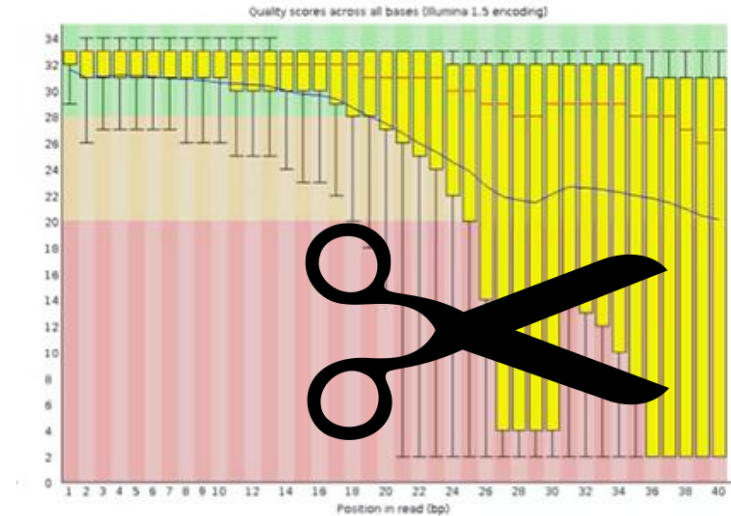
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+
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.....
```



Quality control

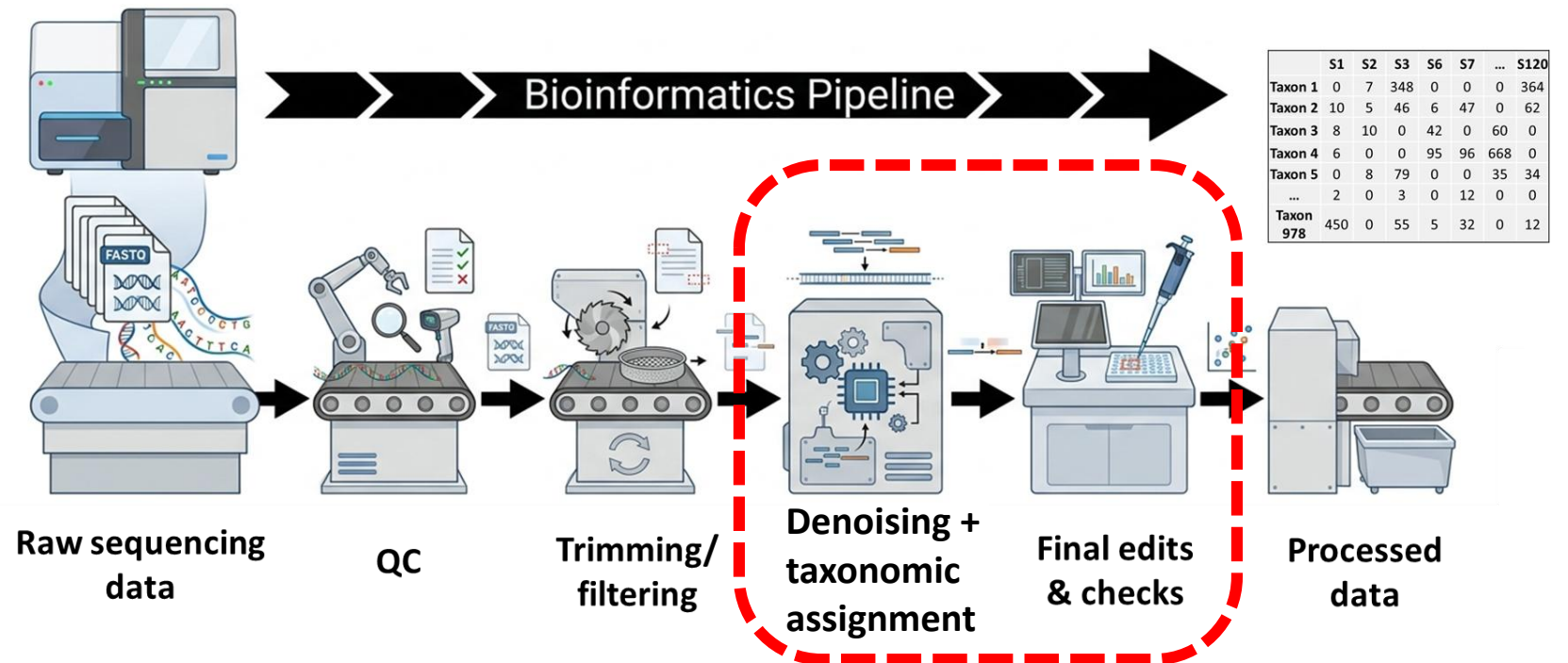
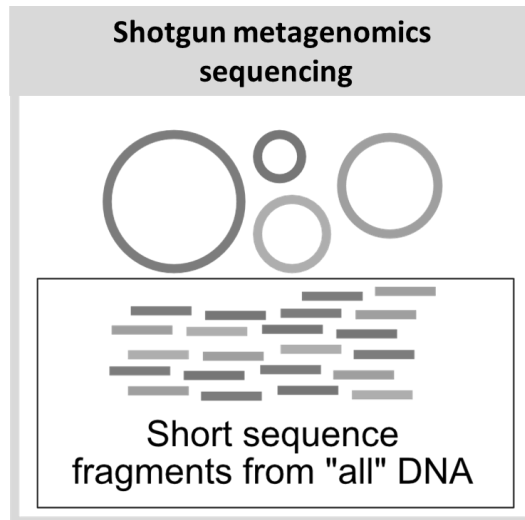
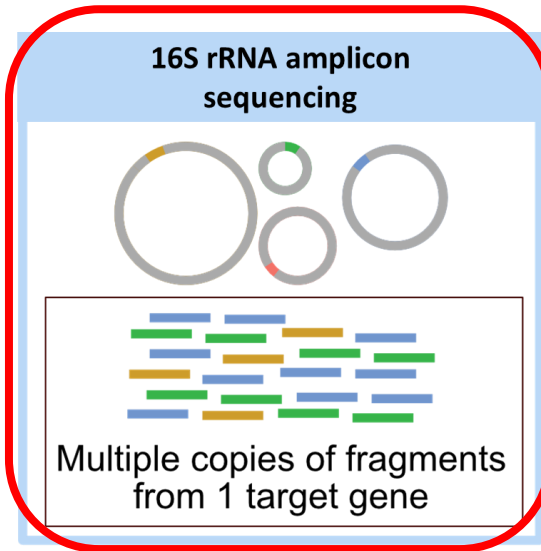
We have reads, but are they trustworthy?

- Low-quality reads
- Sequencing artifacts
- Host DNA contamination
- Other ...



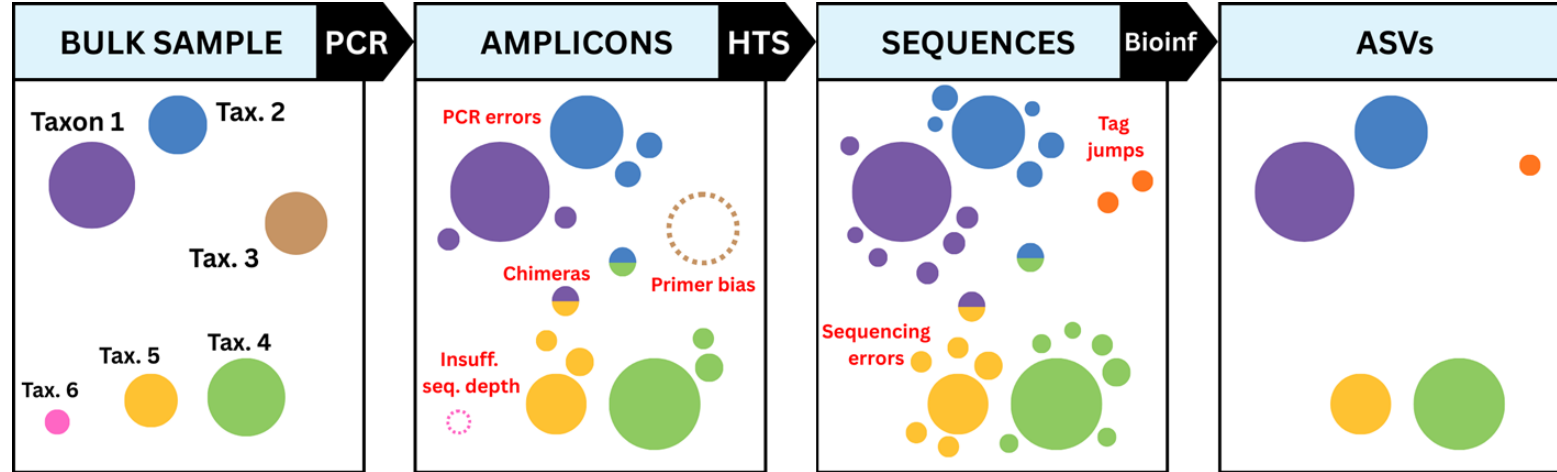
Basics of the bioinformatics pipeline

16S rRNA amplicon data



Is this really a different microbe, or just a typo?

16S rRNA amplicon workflow

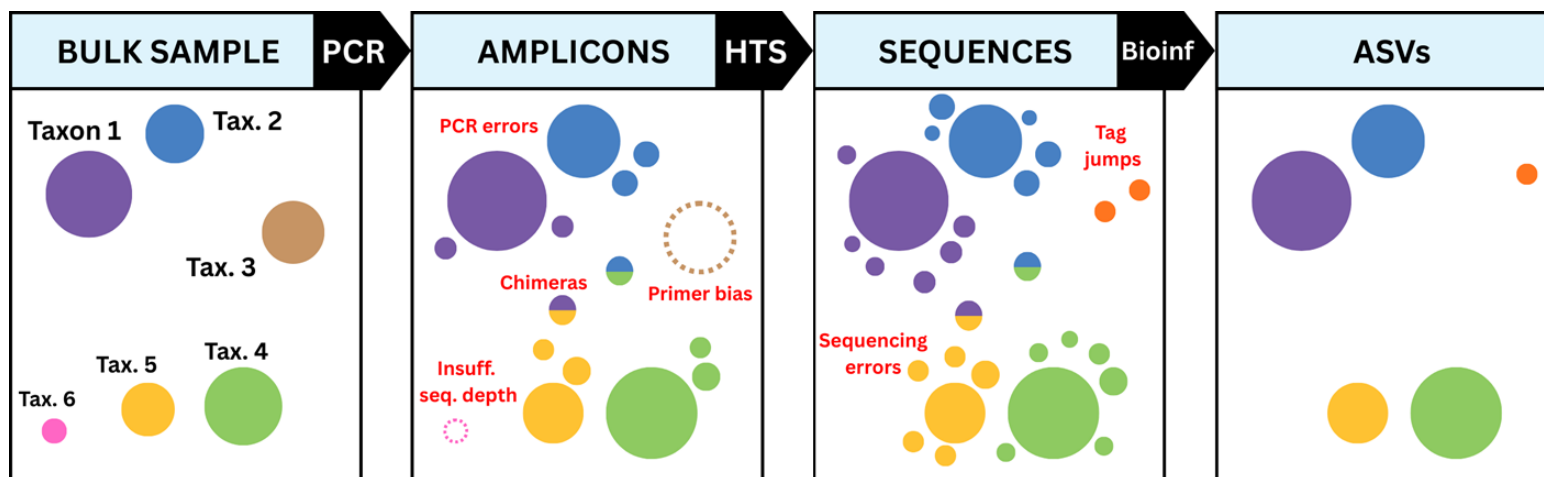


Problem:

- PCR errors
- Sequencing errors
- Chimeras

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16S rRNA amplicon workflow



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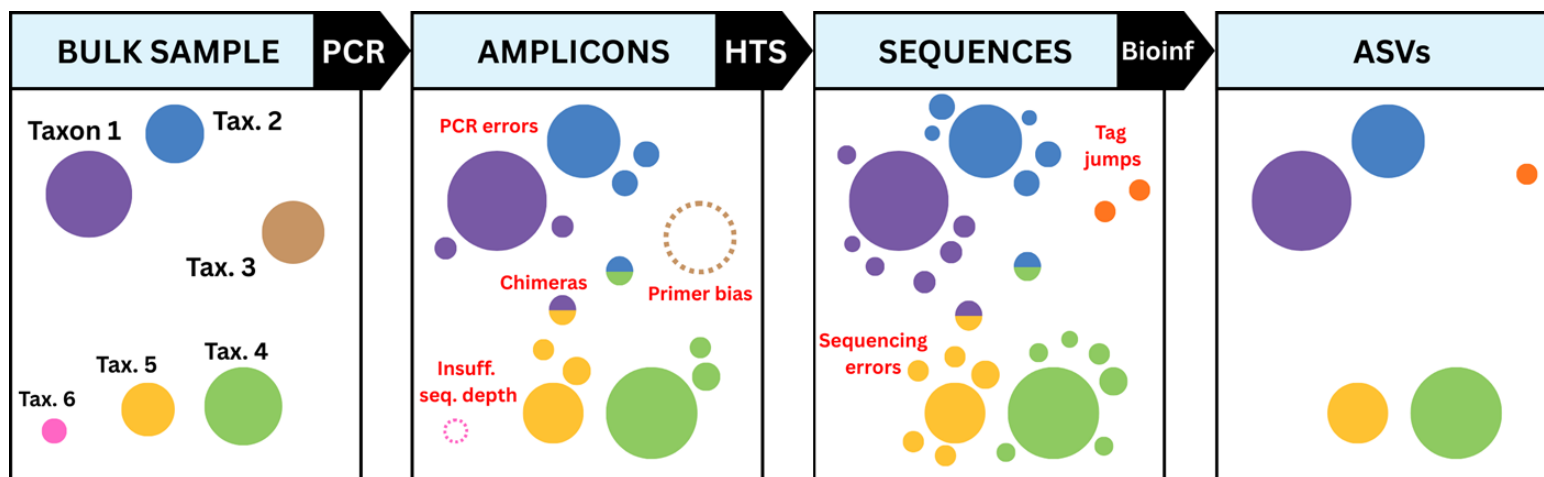
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Consequences:

- Inaccurate taxonomic identification
- Overestimated diversity

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16S rRNA amplicon workflow



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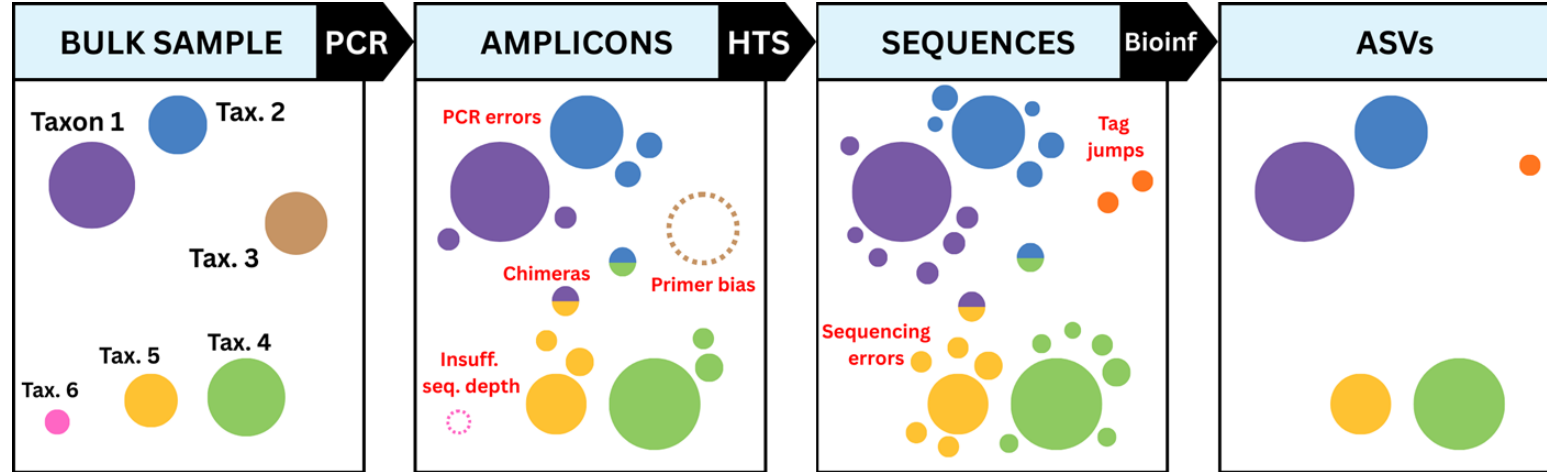
- Inaccurate taxonomic identification
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Solution: denoising

- Corrects sequencer errors
- Removes chimeras
- Corrects point errors from PCR

Is this really a different microbe, or just a typo?

16S rRNA amplicon workflow



Problem:

- PCR errors
- Sequencing errors
- Chimeras

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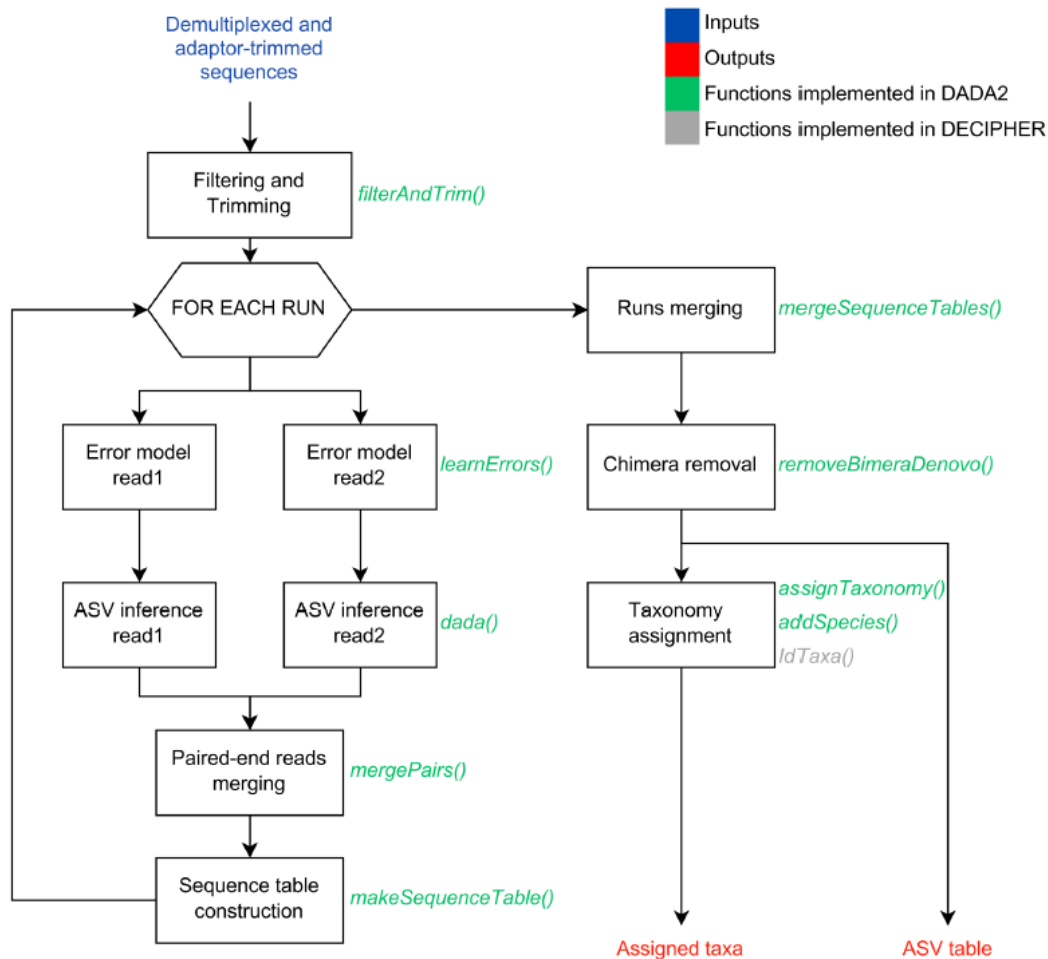
- Corrects sequencer errors
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- Corrects point errors from PCR

TCTG**G**AGGAATAC**C**GGTGGC → TCTG**A**AGGAATAC**C**GGTGGC

Denoising

DADA2

16S rRNA amplicon workflow



1. It learns what errors look like in your run.
2. It then separates real biology from noise, base by base.
3. It finishes by merging, tabulating, and removing chimeras.

Tutorial link:

<https://benjjneb.github.io/dada2/tutorial.html>

Other tools:
DEBLUR, UNOISE3

Okay, we have a clean sequence — but *who* is it?

16S rRNA amplicon workflow

To do taxonomic assignment, you will need:

- a classifier
- a database

TCTG**G**AGGAATAC**C**GGTGGC → TCTG**A**AGGAATAC**C**GGTGGC → ESCHERICHIA

Taxonomic classifier

16S rRNA amplicon workflow

Based on:

A) K-mers

B) Sequence-identity

TCTG**G**AGGAATAC**C**GGTGGC → TCTG**A**AGGAATAC**C**GGTGGC → ESCHERICHIA

Taxonomic classifier

16S rRNA amplicon workflow

Based on:

A) K-mers

TTCTTCGGTT	3-mer	Count
TTC	TTC	2
TCT	TCT	1
CTT	CTT	1
TTC	CTT	1
CCG	CCG	1
CGG	CGG	1
GGT	GGT	1
GTT	GTT	1

Examples:

- assignTaxonomy
- idtaxa
- RDP classifier
- Qiime feature-classifier
classify-sklearn
- SINTAX
- KRAKEN2
- SPINGO

B) Sequence-identity

Color key for alignment scores



Examples:

- Qiime feature-classifier consensus-blast
- Qiime feature-classifier consensus-vsearch
- addSpecies
- BLAST

TCTG**G**AGGAATAC**C**GGTGGC → TCTG**A**AGGAATAC**C**GGTGGC → ESCHERICHIA

Taxonomy database

16S rRNA amplicon workflow

Database	Link
SILVA	https://www.arb-silva.de/
GREENGENES2	https://ftp.microbio.me/greengenes_release/
RDP	https://sourceforge.net/projects/rdp-classifier/
GTDB	https://gtdb.ecogenomic.org/

TCTG**G**AGGAATAC**C**GGTGGC → TCTG**A**AGGAATAC**C**GGTGGC → ESCHERICHIA

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Quick demonstration

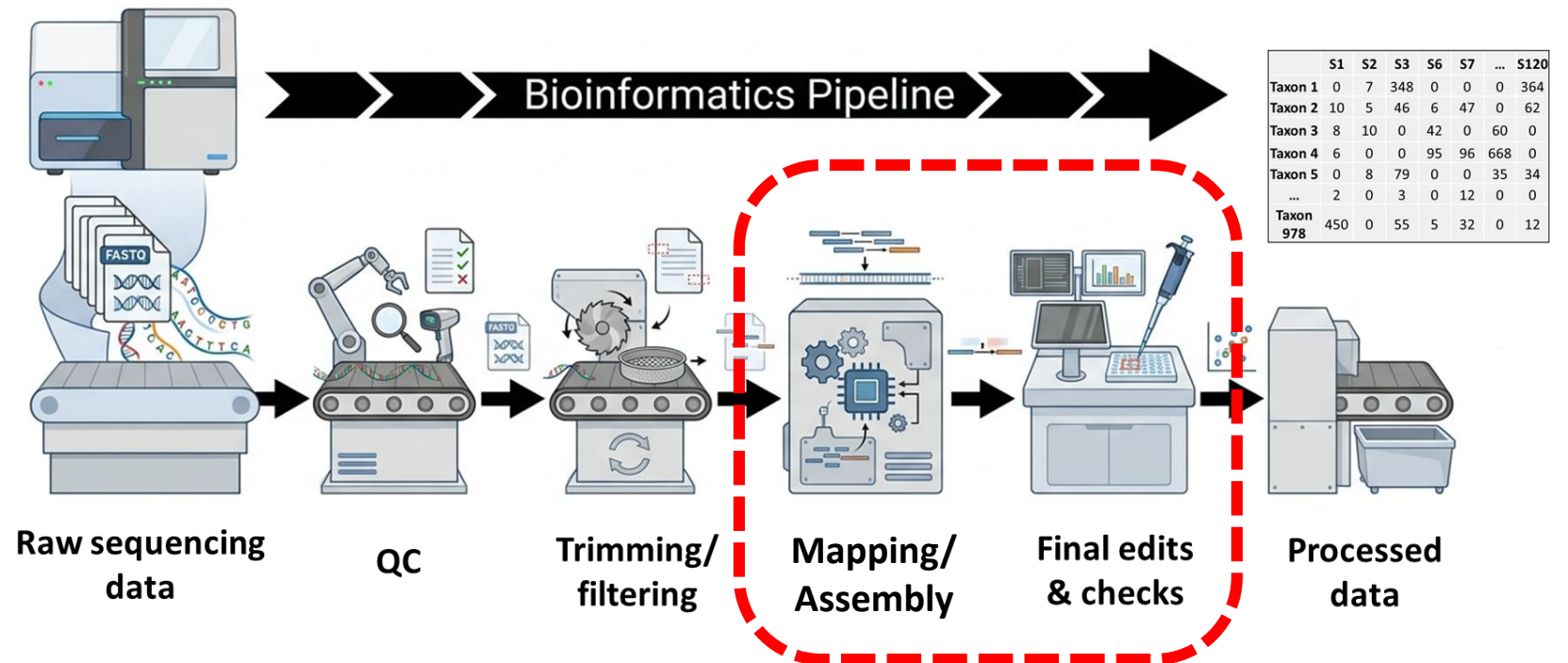
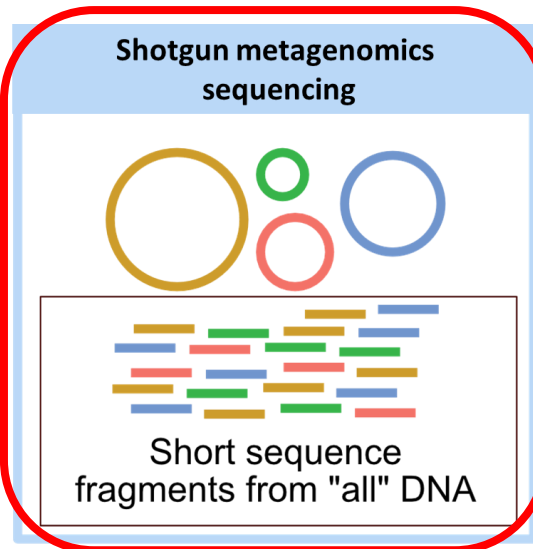
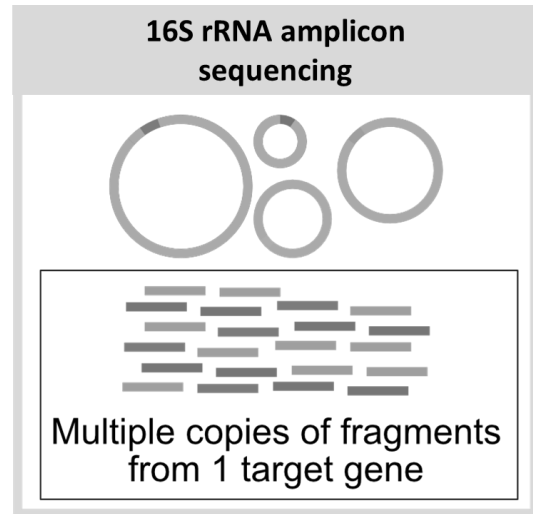
<https://usegalaxy.org/workflows/invocations/535a1685926a8599>

What we'll cover

1. What data does a bioinformatician work with
2. Methodological comparison of 16S and shotgun whole-metagenome sequencing
- 3. Basics of the bioinformatics pipeline**
4. Quick demonstration (Galaxy)

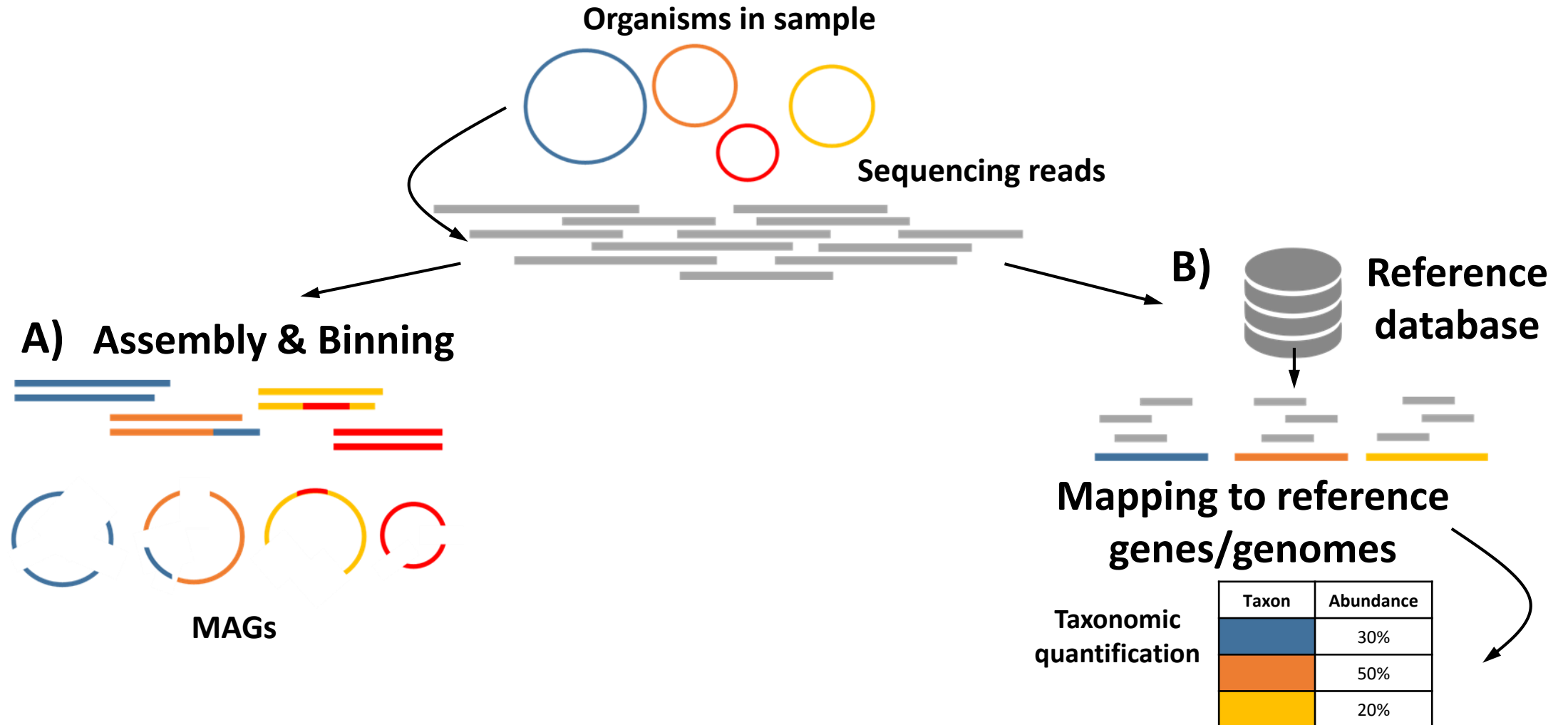
Basics of the bioinformatics pipeline

WMGS



Basics of the bioinformatics pipeline

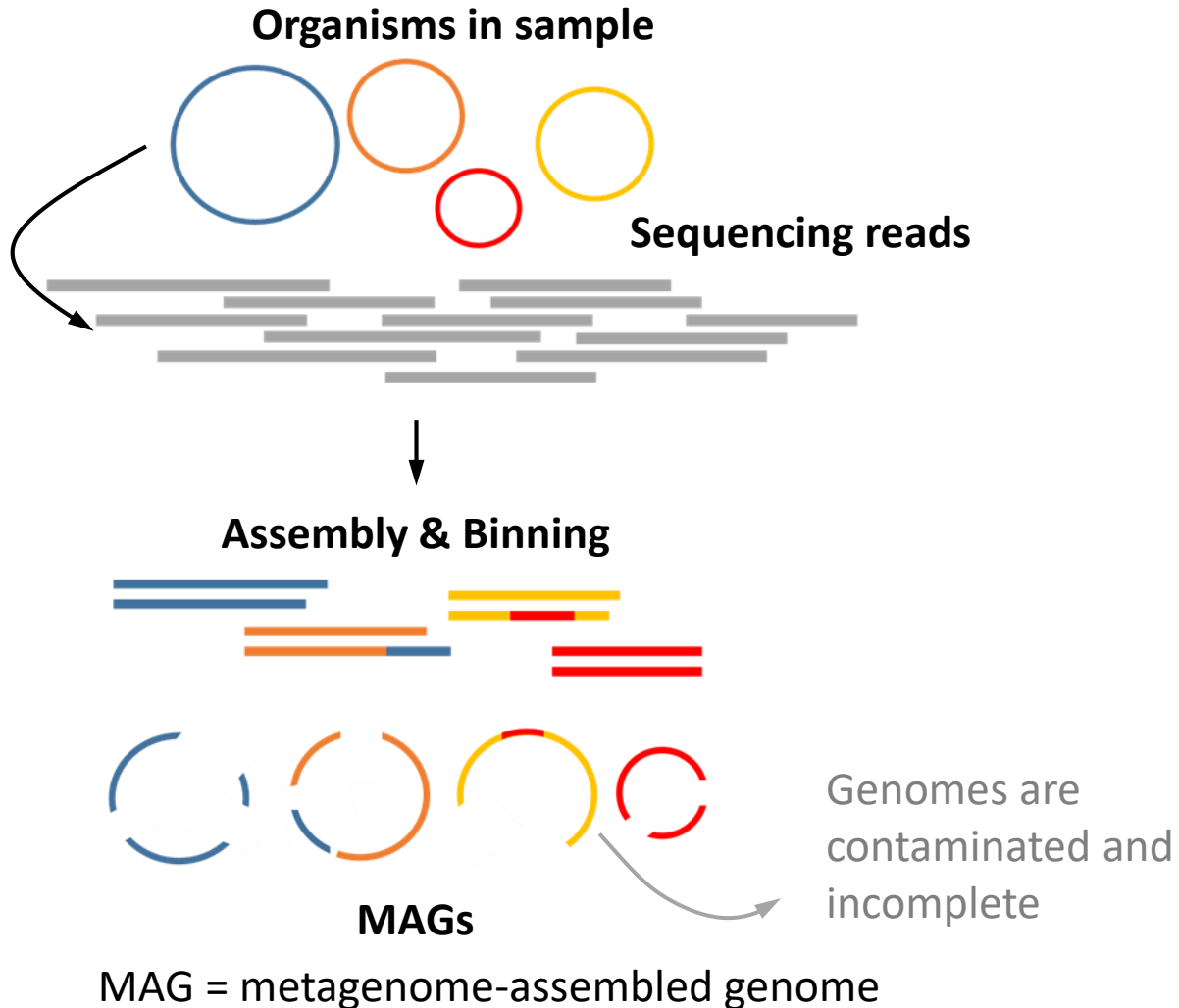
WMGS



Basics of the bioinformatics pipeline

WMGS

A)

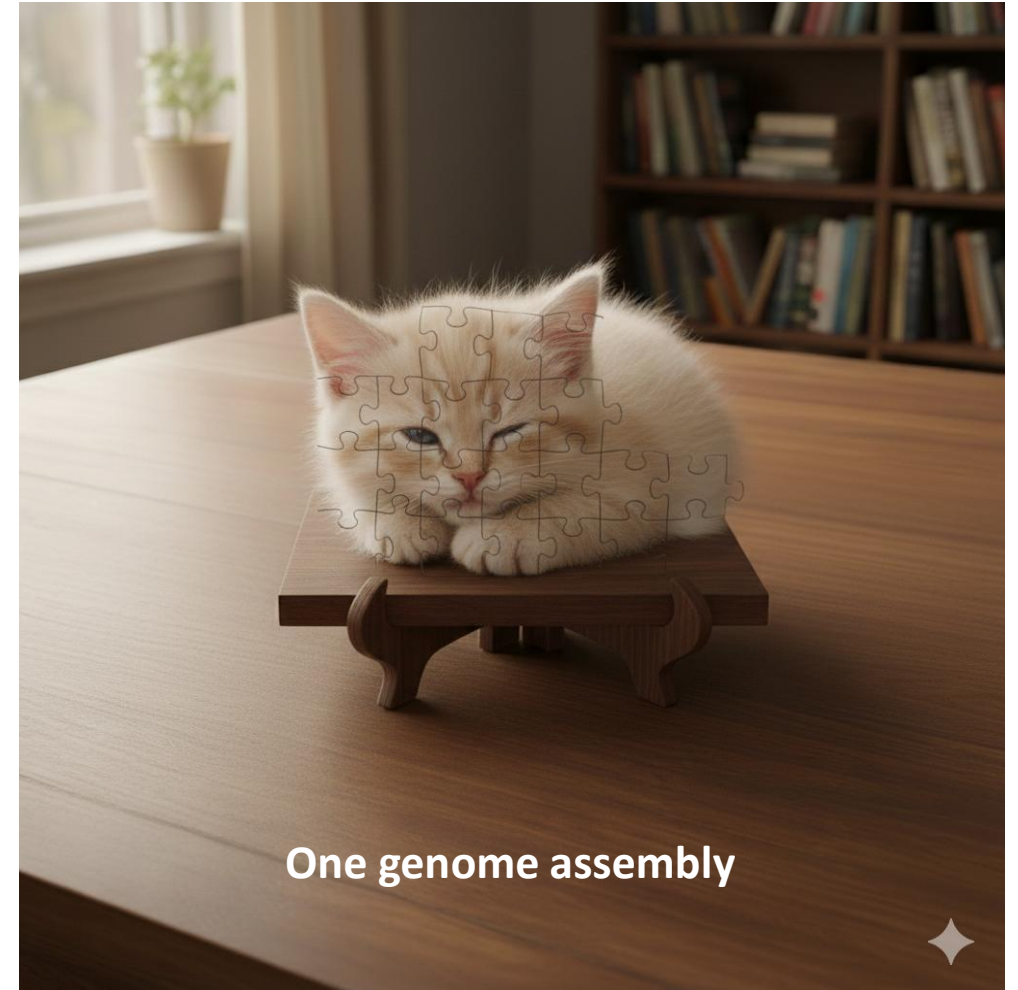
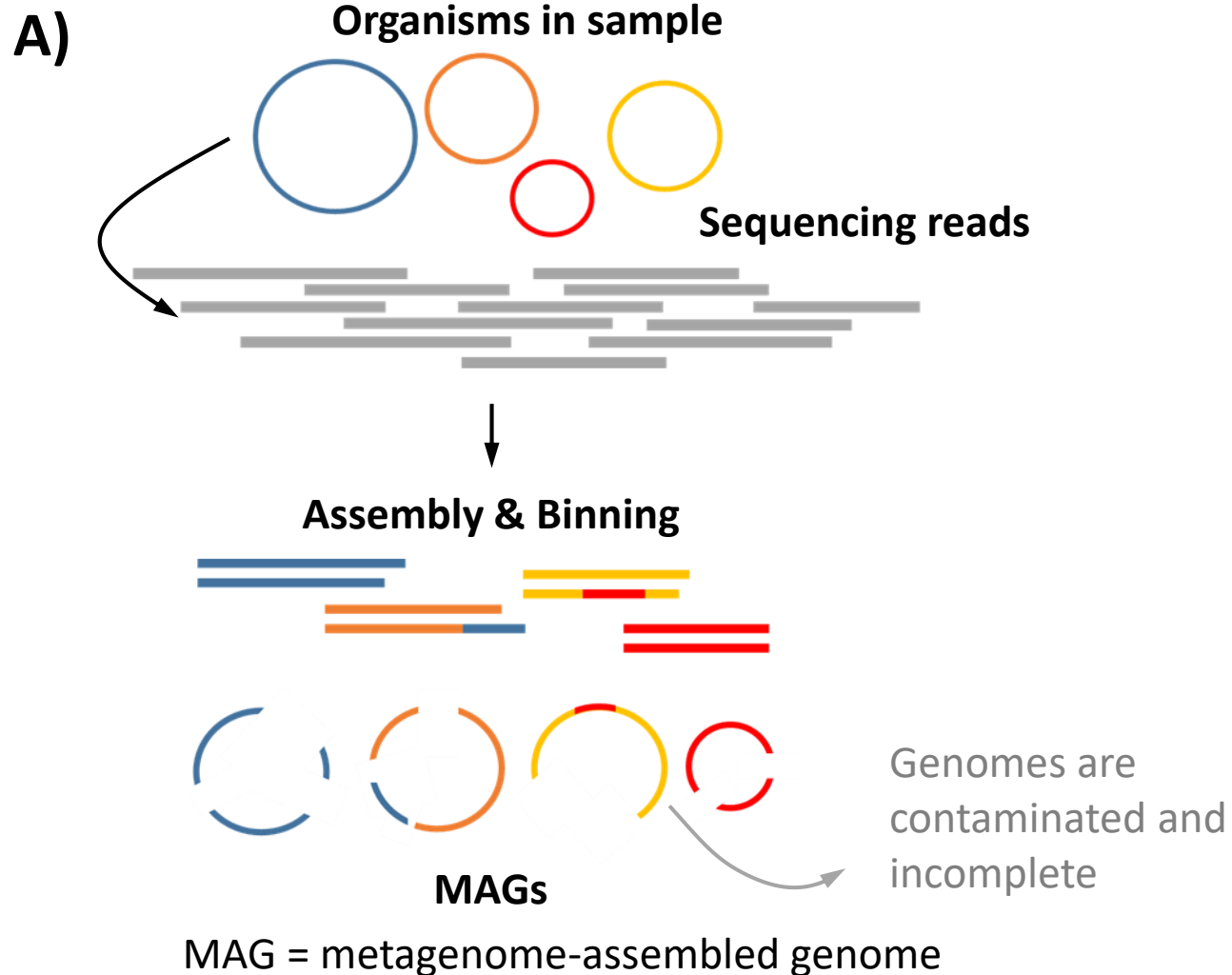


Examples:

MEGAHIT, metaSPAdes, metaFlye, Canu ...
MetaBAT2, MaxBin2, GTDB-Tk

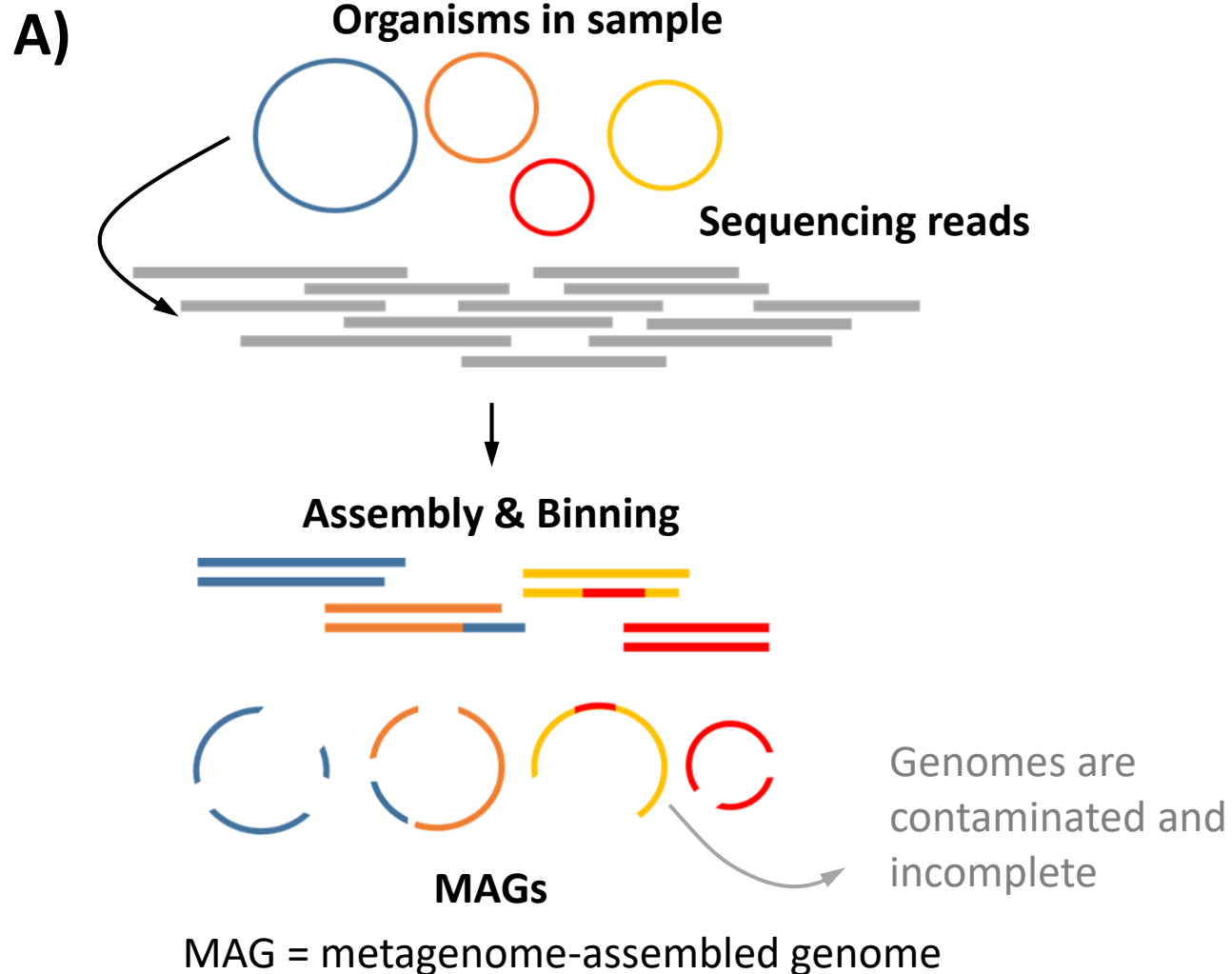
Basics of the bioinformatics pipeline

WMGS



Basics of the bioinformatics pipeline

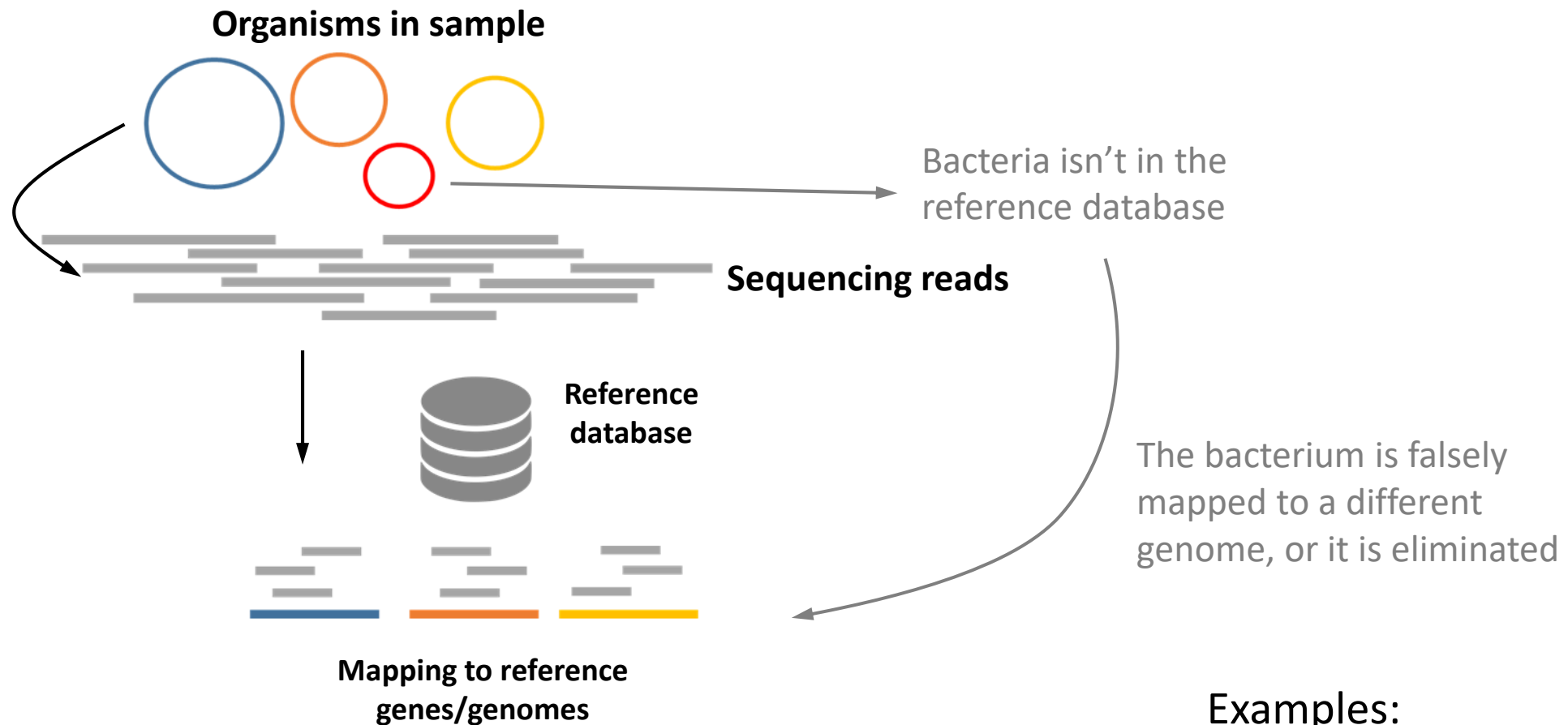
WMGS



Basics of the bioinformatics pipeline

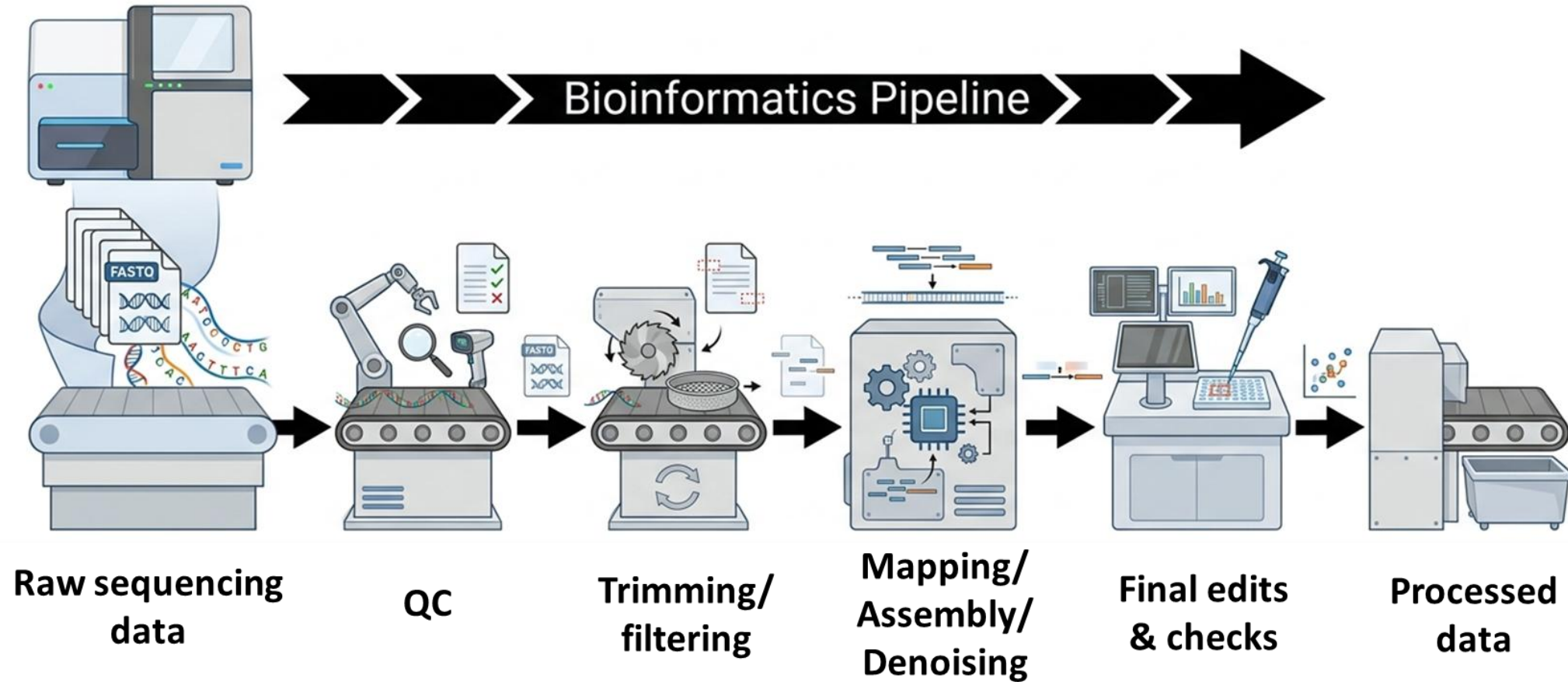
WMGS

B)



Examples:
Kraken, mOTU, MetaPhlan ...

From machine output to biology



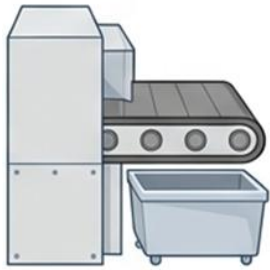
From machine output to biology

Count table

TaxonID	S1	S2	S3	S6	S7	...	S120
Taxon 1	0	7	348	0	0	0	364
Taxon 2	10	5	46	6	47	0	62
Taxon 3	8	10	0	42	0	60	0
Taxon 4	6	0	0	95	96	668	0
Taxon 5	0	8	79	0	0	35	34
...	2	0	3	0	12	0	0
Taxon 978	450	0	55	5	32	0	12

Taxonomic table

TaxonID	Kingdom	Phylum	Class	Order	Family	Genus
Taxon 1	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Bacteroidaceae	Bacteroides
Taxon 2	Bacteria	Bacteroidota	Bacteroidia	Bacteroidales	Prevotellaceae	Prevotella
Taxon 3	Bacteria	Bacillota	Clostridia	Lachnospirales	Lachnospiraceae	Roseburia
Taxon 4	Bacteria	Bacillota	Clostridia	Oscillospirales	Oscillospiraceae	Faecalibacterium
Taxon 5	Bacteria	Bacillota	Bacilli	Lactobacillales	Lactobacillaceae	Lactobacillus
...	Bacteria	Verrucomicrobiota	Verrucomicrobiia	Verrucomicrobiales	Akkermansiaceae	Akkermansia
Taxon 978	Archaea	Methanobacteriota	Methanobacteria	Methanobacteriales	Methanobacteriaceae	Methanobrevibacter



Processed
data

From machine output to biology

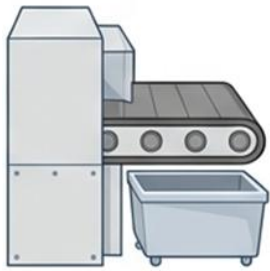
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This is where
bioinformatics ends
and your biology
begins

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Taxon 978	Archaea	Methanobacteriota	Methanobacteria	Methanobacteriales	Methanobacteriaceae	Methanobrevibacter



Processed
data

Thank you for your attention!

Questions?



Petra Polakovicova
September 2026